

FIGURE 2.1: *Early photos of me as an infant and toddler. In each photo, I am looking out of one eye while the other eye is turned in. (Barry family photos)*

During each visit, Dr. Fasanella had me look at a penlight with my left eye while he covered my right eye with an occluder, a tool that looked like a large, flattened spoon. While I was looking at the penlight, he moved the occluder from my right to left eye and observed that my right eye straightened to look at the penlight while my left eye, behind the occluder, turned in. If he covered the right eye again, then the left eye moved out to look at the penlight while the right eye behind the occluder turned in. I could straighten whichever eye was doing the looking. Like most cross-eyed children, I had a nonparalytic form of strabismus: I had no problem moving each eye in its orbit. My muscles were functioning fine, but the coordination of the two eyes was off.

My parents were relieved to have a definite diagnosis and asked Dr. Fasanella if problems with my vision explained why I was such a difficult toddler, prone to crying and pounding my head against the floor. He told them that my depth perception was poor—but added nothing further. Doctors knew much less about vision in the 1950s than they do now, and the idea that crossed eyes and other binocular disorders might affect child development wasn't given serious consideration. Had I been born today, my parents may have received more detailed answers.



Babies are born with their eyes wide open, but what do they see? The great psychologist William James wrote that infants are born into “blooming, buzzing confusion.” However, research on infants indicates that the world may not be as confusing to infants as James imagined. A baby’s sensory world is actually very different from an adult’s. At birth, the fovea of the retina is not fully developed. Since this area is responsible for our sharpest vision, infants don’t see with the kind of clarity experienced by an adult. Newborn babies do not have the ability to focus their eyes at different viewing

distances. Instead, they see well only at a distance of about nine inches, the right distance to see their mother's face while nursing. But newborns have some innate perceptual skills—babies, at just nine minutes old, exhibit a preference for looking at a human face.

In addition, the eyes of a very young infant are not always straight. They may wander out of alignment several times a day, but these misalignments do not usually lead to strabismus. A habitual crossing of the eyes emerges at its earliest at about two to three months of age. The early misalignments may result from the baby's attempts to move her eyes into position for viewing at different distances, not just at the distance of her mother's face while nursing. Infants start to look at near objects by converging and then at far objects by diverging their eyes. Most demonstrate an ability to make these vergence movements by about twelve weeks.

During this same period, the fovea of the eye is maturing to allow for sharper vision, and the child develops the ability to focus the lens of the eye in order to see clearly both far and near.

Doctors can watch how babies move their eyes, but it is much more difficult to determine what an infant actually sees. Does a baby see in 3D in the first few weeks of life? Since the eyes are separated in space, the view from each eye is slightly different, and the brain takes advantage of this difference to allow us to see with stereopsis. So, to develop stereopsis, a baby's brain must compare the images seen by the right and left eye. The brain needs to know which eye is seeing what.

One way to determine what babies see is to study what babies like to look at. With these "preferential-looking" experiments, scientists have learned when an infant can see in 3D. To establish when the brain starts comparing the images from the two eyes, scientists at the Massachusetts Institute of Technology (MIT) placed Polaroid goggles on healthy babies whose parents had agreed to have them participate in experiments. These goggles were very much like the Polaroid glasses used in 3D movie theaters. The babies were then

## Notes

## **Chapter 1: Stereoblind**

***1 He was describing the development of the visual system, highlighting experiments done on walleyed and cross-eyed kittens.***

Hubel DH, Wiesel TN. Binocular interaction in striate cortex of kittens reared with artificial squint. *Journal of Neurophysiology* 28 (1965): 1041-59.

Hubel DH, Wiesel TN. *Brain and Visual Perception: The Story of a 25-Year Collaboration*. Oxford: Oxford University Press, 2005.

***2 We have two eyes, he said, but only one view of the world.***

An excellent review of stereovision is found in Hubel DH. *Eye, Brain, and Vision*. New York: Scientific American Library, 1995. Also available online at <http://hubel.med.harvard.edu/bcontex.htm>.

***3 The classroom didn't seem entirely flat to me.***

See chapter 7 for a more detailed discussion of nonstereoscopic cues to depth.

***4 Many of the great students of optics, including Euclid, Archimedes, da Vinci, Newton, and Goethe, never figured out how we see in stereoscopic depth***

For a fascinating review of the history of our knowledge of binocular vision, see

Howard IP, Rogers BJ. *Seeing in Depth*. Ontario: I. Porteus, 2002, vol. 1, ch. 2.

Crone RA. *Seeing Space*. Exton, PA: Swets and Zeitlinger Pubs., 2003.

***6 The teddy bear located to your left casts its image on corresponding points on the right side of both your retinas, while the rattle, to the right, casts its image on corresponding points on the left side of both retinas.***

In [figure 1.3](#), the block is located at the fixation point. The bear, block, and rattle all fall on a line called the horopter, or, more specifically, the apparent frontoparallel plane horopter. Although this line is slightly curved, we interpret the bear, the rattle, and all other objects along this horopter as located in the same plane as the block. See, for example,

Steinman SB, Steinman BA, Garzia FP. *Foundations of Binocular Vision: A Clinical Perspective*. New York: McGraw-Hill Cos., 2000, ch. 4.

**6 In 1838, Wheatstone explained how the relative position of the images on the two retinas allows us to see in 3D.**

Wheatstone C. Contributions to the Physiology of Vision.—Part the First. On some remarkable, and hitherto unobserved, Phenomena of Binocular Vision. *Philosophical Transactions of the Royal Society of London* 128 (1838): 371-94.

**10 I thought about people who were totally colorblind.**

Individuals with a hereditary form of total colorblindness are described in

Sacks O. *The Island of the Colorblind*. New York: Alfred A. Knopf, 1996.

**10 With this knowledge, could they see in their mind's eye what they couldn't see in the real world?**

Many philosophers have pondered this question. For example, Frank Jackson posed the famous thought experiment about Mary the Neuroscientist. Mary is brilliant and knows everything there is to know theoretically about color and color vision. However, she has lived all her life in a black-and-white room, her entire body covered in black-and-white clothes, so that she saw absolutely no color. Finally, Mary is let out of her room. She sees red for the first time. Is red what she imagined? Had she been able to imagine any of the colors that she now sees? Has she learned something new about the world?

Jackson F. What Mary didn't know. *Journal of Philosophy* 83 (1986): 291-95.

Also see Ludlow P, Nagasawa Y, Stoljar D (eds.). *There's Something about Mary: Essays on Phenomenal Consciousness and Frank Jackson's Knowledge Argument*. Cambridge, MA: MIT Press, 2004.

Thanks to Greg Frost-Arnold for introducing me to Mary.

***11 If you learn to read braille even as an adult, the number of neurons in your brain that receive touch input from your reading index finger increases.***

Pascual-Leone A, Torres F. Plasticity of the sensorimotor cortex representation of the reading finger of braille readers. *Brain* 116 (1993): 39-52.

***11 In the 1990s, scientists studied the brains of violinists with magnetic source imaging, and they found that more neurons in the motor cortex of violinists were devoted to the control of the fingers of the left than the right hand.***

Elbert T, Pantev C, Wienbruch C, Rockstroh B, Taub E. Increased cortical representation of the fingers of the left hand in string players. *Science* 270 (1995): 305-7.

***13 You might think that Jenny's experiments indicated that Dan's ability to sense and move had degenerated while he was in space. Instead, Dan had adapted to a radically new environment, the microgravity of outer space.***

Reschke MF, Bloomberg JJ, Harm DL, Paloski WH. Space flight and neurovestibular adaptation. *Journal of Clinical Pharmacology* 34 (1994): 609-17.

Reschke MF, Bloomberg JJ, Harm DL, Paloski WH, Layne C, McDonald V. Posture, locomotion, spatial orientation, and motion sickness as a function of space flight. *Brain Research Reviews* 28 (1998): 102-17.

***14 Since my misaligned eyes saw different things, they competed for input onto visual cortical neurons, and on each neuron, one or the other eye won out.***

Hubel DH, Wiesel TN. Binocular interaction in striate cortex of kittens reared with artificial squint. *Journal of*